

A torsion $\mathbb{Z}/22$ curve over a quartic field

Problèmes ouverts — A. Dujella · 2026

Problem 2.9 (Dujella, 2026). Is there an elliptic curve over a quartic field with torsion group $\mathbb{Z}/22\mathbb{Z}$ and positive rank?

Remark. Solved by Maksym Voznyy on 2024/01/04. There is a curve with torsion group $\mathbb{Z}/22\mathbb{Z}$ and rank 2 over the quartic field defined by the polynomial $f(x) = x^4 + \frac{50}{27}x^3 + \frac{281}{243}x^2 + \frac{68}{243}x + \frac{2}{81}$. The curve has the equation

$$y^2 - (201204z^3 + 328374z^2 + 159732z + 8019)xy - (14996874942306z^3 + 23680026731268z^2 + 10746821159640z + 93126) \\ = x^3 - (1435244994z^3 + 2266248132z^2 + 1028502360z + 89124624)x^2,$$

where z is a root of f .